

To: PhD Admissions Committee, PhD in Computer Science (XLII cycle) — Sapienza University of Rome **Attn:** Prof. Roberto Navigli, Sapienza NLP group **Subject:** Motivation letter — PhD in Computer Science (XLII / AY 2026-27) — Multilingual NLP / LLM Interpretability — Nauval Zulfikar

Dear Members of the Admissions Committee, dear Prof. Navigli,

I am applying to the PhD in Computer Science at Sapienza to work, within the Sapienza NLP group, on the interpretability of multilingual large language models — specifically, how lexical-semantic knowledge is represented inside these models and transferred across languages. The group's combined depth in multilingual NLP, lexical semantics, and large language models such as the Minerva Italian-English family is, as far as I have found, an unusually precise match for the questions I most want to pursue.

My route into this is through transformer representations of meaning. For my MSc dissertation at Aston University (Business Analytics, First Class), I fine-tuned DeBERTa-v3 for multi-aspect classification on multi-tier supplier-review text and evaluated its emergent biases against literature baselines, embedding the model inside a larger decision system. In a related project, my CV-JD Suitability Checker used DeBERTa-v3 semantic matching to outperform lexical ATS baselines — concrete evidence that I work natively with how transformers encode meaning rather than surface form. My LLM-Shipper-Profiles system, a retrieval-augmented gpt-4o-mini pipeline, was built deliberately to be reproducible and ablatable, because I believe claims about model behaviour are only worth as much as the experiments that can be re-run to check them.

That same discipline — ablation and reproducibility as defaults, per-class rather than aggregate-only evaluation — is what I would bring to interpretability research. My proposed project asks where, layer by layer, a multilingual model's representation of a concept becomes language-agnostic, and which internal components causally mediate cross-lingual semantic transfer, using word-sense disambiguation as a grounded probe over the group's multilingual semantic resources.

I am candid that my degree is titled Business Analytics rather than Computer Science; my portfolio of transformer fine-tuning, RAG, and controlled experimentation is offered as direct evidence of the CS competence the program expects, and I am eager to deepen the theoretical foundations under expert supervision. As my MSc at Aston was taught and assessed in English, I expect to qualify for the English-language exemption (IELTS 7.0 overall available as backup).

It would be a privilege to pursue this work in your group. Thank you for considering my application.

Sincerely, Nauval Zulfikar zulfikar.nauval1998@gmail.com | +44 7300 469048 | nauvalzulfikar.vercel.app

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